上海交通大学学位论文

# 面向 3C 工业生产线的双臂机器人视觉-语言-动作模型


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# A VISION-LANGUAGE-ACTION MODEL FOR DUAL-ARM ROBOTS IN 3C INDUSTRIAL PRODUCTION LINES


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# 上海交通大学

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# 摘　要

学位论文是本科生从事科研工作的成果的主要表现，集中表明了作者在研究工作中获得的新的发明、理论或见解，也是科研领域中的重要文献资料和社会的宝贵财富。

为了提高本科生学位论文的质量，做到学位论文在内容和格式上的规范化与统一化，特制作本模板。

**关键词：** 双臂机器人，视觉-语言-动作模型，工业自动化

# ABSTRACT

As a primary means of demonstrating research findings for undergraduate students, dissertation is a systematic and standardized record of the new inventions, theories or insights obtained by the author in the research work. It can not only function as an important reference when students pursue further studies, but also contribute to scientific research and social development.

This template is therefore made to improve the quality of undergraduates' dissertation and to further standardize it both in content and in format.

**Key words:** dual-arm robot, vision-language-action model, industrial automation

# Contents

# Chapter 1 Introduction

## 1.1 Background and significance

This chapter introduces the project background, motivation, and overall scope of developing a vision-language-action model for dual-arm robots in 3C industrial production lines.

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## 1.2 Project objectives and scope

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## 1.3 Organization of this thesis

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## 2.2 Engineering specifications

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## 3.1 Concept generation

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**Table 2–1 Bill of materials**

| Quantity | Part Description | Purchased From | Part Number | Price (each) | Notes |
|---|---|---|---|---|---|
| ... | ... | ... | ... | ... | ... |

# Research Projects and Publications during Undergraduate Period

[1] ……

# Acknowledgements

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# A VISION-LANGUAGE-ACTION MODEL FOR DUAL-ARM ROBOTS IN 3C INDUSTRIAL PRODUCTION LINES

HCCI (Homogenous Charge Compression Ignition) combustion has advantages in terms of efficiency and reduced emission. HCCI combustion can not only ensure both the high economic and dynamic quality of the engine, but also efficiently reduce the NOx and smoke emission. Moreover, one of the remarkable characteristics of HCCI combustion is that the ignition and combustion process are controlled by the chemical kinetics, so the HCCI ignition time can vary significantly with the changes of engine configuration parameters and operating conditions. ⋯⋯